PropValue Real Estate Analytics

Week 1 Milestone Report | Stakeholder Brief & Data Inventory | Revised after AI review

1. Project objective

Develop a real-estate analytics platform that compares property markets across three micro-markets, examines price trends, estimates fair property values, and supports an interactive map dashboard and valuation API. The Week 1 objective is to clarify stakeholder questions and establish a reliable inventory of the data required for the analysis.

2. Key analytical questions

1. How do listing prices and price per square foot change over time in each of the three selected micro-markets?

2. How do median price and median price per square foot differ between the three micro-markets, after accounting for property type and size where possible?

3. Which listing attributes (such as area, bedrooms, property type, property age, and location, if available) are most associated with listing price?

4. How much variation exists in price and price per square foot within each micro-market, and are there suspicious outliers or inconsistent records?

5. How accurately can a baseline model forecast prices for a future time period, measured using MAPE on a time-based holdout set?

3. Data inventory status

The real-estate listings dataset is the critical source for the project, but its download link returned an error and the file could not be inspected. The two accessible CSV files were profiled as a preliminary exercise; their relevance to the real-estate objective has not been established.

File	Observed structure	Quality checks	Relevance / status
da_realestate_listings.csv	Not inspected; file unavailable	Cannot assess until file is obtained	Required primary dataset; download link returned 404 / redirects to login
da_customer_profiles.csv	200 rows × 10 columns	0 missing cells; 0 exact duplicate rows	Retail/customer-style data; relevance to property analysis unconfirmed
da_product_catalogue.csv	50 rows × 7 columns	0 missing cells; 0 exact duplicate rows	Product catalogue data; relevance to property analysis unconfirmed

4. Data quality assessment plan

Once the real-estate listings dataset is available, the following checks will be completed:

- Confirm row and column counts, column names, data types, and date coverage.
- Check missing values, exact duplicates, and duplicate property/listing identifiers.
- Validate price, area, bedrooms, dates, and location fields for valid ranges and formats.
- Check whether the three micro-markets can be identified consistently.
- Review price-per-square-foot calculations and investigate outliers or inconsistent records.
- Document limitations, assumptions, and any data-cleaning decisions.

5. Stakeholder clarification questions

• Which three micro-markets should be included, and what boundaries define each market?
• What is the preferred definition of fair value: listing price, transaction price, or another measure?
• What time period and forecast horizon should the price-trend model cover?
• Is there a preferred source of property data, and are listing prices verified or asking prices?
• What format and performance requirements are expected for the valuation API and interactive map?

6. Limitations and next steps

The current inventory is preliminary. The two accessible CSVs have been checked for basic completeness and exact duplicates, but they do not establish that suitable property-level data is available. No real-estate price analysis or forecasting model has been performed.

Next steps: request a working copy of da_realestate_listings.csv from Cynaris support; inspect and profile its actual columns and values; confirm market definitions with the stakeholder; then finalize the data inventory and proceed to exploratory analysis.

Integrity note: This report distinguishes completed preliminary checks from work blocked by the missing primary dataset. No property data or analytical results have been fabricated.